Data and Information Management 8 (2024) 100053

Contents lists available at ScienceDirect

Data and Information Management

journal homepage: www.journals.elsevier.com/data-and-information-management

Digital platform ecosystem governance of private companies: Building blocks and a research agenda based on a multidisciplinary, systematic literature review

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ARTICLE INFO

Keywords:
Digital platform ecosystem governance
Systematic literature review
Conceptual model
Research agenda

ABSTRACT

Digital platform ecosystem governance refers to a platform owner's decisions and mechanisms that seek to influence complementors and users to build and sustain an ecosystem. The relevance of digital platform ecosystem governance is broadly acknowledged by researchers. However, the extant body of research is fragmented, and varied terminologies are employed, leading to challenges in identifying and recognizing results across different fields. This article provides a multidisciplinary and systematic literature review with the aim of consolidating knowledge on this important topic. Based on an analysis of 103 journal articles and conference papers, this review synthesizes the literature into a conceptual model with five building blocks of platform ecosystem governance. The model aims to create a robust foundation for researchers approaching the topic for the first time and conducting subsequent research. The conceptual model also offers practical guidance for governing ecosystems in a structured manner. Finally, this article provides a research agenda with five areas for future investigation.

1. Introduction

"Digital platform ecosystem governance" is understood as a platform owner's decisions and mechanisms that seek to influence complementors and users to build and sustain an ecosystem. The term refers to orchestration rather than coercion because digital platform ecosystems are more structured than a market but less structured than a supply chain (Foss et al., 2023; Jacobides et al., 2018; Parker & Van Alstyne, 2018, as cited in Halckenhaeusser et al., 2020). Adopting a sociotechnical view, digital platform ecosystems can be defined as evolving meta-organizations consisting of a platform and all the actors interacting on that platform, which are coordinated through means other than a hierarchy (Gawer, 2014; Gulati et al., 2012; Jacobides et al., 2018). Without the possibility of enforcing direct control of a multitude of autonomous actors, platform governance aims to address the delicate balance between retaining and relinquishing control to ensure that actors will join, use, and enhance the platform's purpose while maintaining its integrity (Gawer, 2014; Tiwana et al., 2010). Governance influences the evolution of a digital platform ecosystem, which can make a huge difference in a business landscape characterized by increased competition across digital platform ecosystems (Tiwana et al., 2010). Indeed, around one out of three platform ecosystems has failed due to poor governance choices (Floetgen et al., 2022; Pidun et al., 2020; Reeves et al., 2019).

Given the importance of this topic at the theoretical and practical levels, research interest is increasing. However, despite this interest, the extant body of knowledge remains fragmented (de Vasconcelos Gomes et al., 2021; Mukhopadhyay & Bouwman, 2019; Vesselkov et al., 2019). The topic has been examined in isolation across different streams of research, resulting in relevant but partial understandings, which can also be attributed to the variegated terminologies used (de Reuver et al., 2018; de Vasconcelos Gomes et al., 2021; Halckenhaeusser et al., 2020; Rietveld & Schilling, 2021). Moreover, research has focused on different aspects, such as pricing in the field of economics and control and the division of decision rights in the field of information systems (Vesselkov et al., 2019). Therefore, there seems to be a need for a review of the literature on digital platform ecosystem governance to synthesize and consolidate extant research across fields and to identify future research avenues. To this end, the present article provides a multidisciplinary, systematic literature review in the fields of information systems (IS), strategy, organization studies, innovation, and economics. This literature review was guided by two research questions: (1) "What are the building blocks of digital platform ecosystem governance?" and (2) "What do we need more research on?"

This review makes two contributions to the literature. First, this review develops a conceptual model of digital platform ecosystem governance by grouping governance concepts into five building blocks

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https://doi.org/10.1016/j.dim.2023.100053
Received 22 May 2023; Received in revised form 27 September 2023; Accepted 5 October 2023
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(governance structure, governance mechanisms, governance scope, contingency factors, and outcomes) and by suggesting how these blocks are interrelated—a topic that has not yet been addressed by extant research (Perscheid et al., 2020; Schreieck et al., 2016). This review also makes the concept of "governance scope" more explicit. The proposed conceptual model aims to present a broad overview of governance, which is relevant for researchers who are approaching the topic for the first time; provide a foundation for further research, such as empirical investigations into the relationships among the building blocks; and assist practitioners in understanding what they need to consider when governing their digital platform ecosystems in a more structured manner. Second, the paper suggests avenues for future research to guide and move our knowledge forward.

The remainder of the paper is structured as follows. First, Section 2 provides a detailed description of the method used in conducting this review. Section 3 describes how the findings are structured according to the conceptual model and the development of the proposed model. Section 4 delineates the findings from the analysis of the extant research. Section 5 discusses the findings and offers a research agenda with questions for future investigation. Finally, Section 6 presents the conclusions and limitations.

2. Method

This review is a stand-alone literature review, as opposed to reviews that aim at setting the theoretical background for a specific research question and researchers' own endeavors (Okoli & Schabram, 2010). Therefore, this article provides a rigorous review of extant research results (Okoli & Schabram, 2010). Based on Fink's (2013) work, this review is *systematic* in following a methodological approach, *explicit* in explaining the procedures used, *comprehensive* by including all relevant materials, and *reproducible* by others who adopt the same approach. This review aims at synthesizing existing literature, identifying gaps in current research, and providing a conceptual framework as a foundation and point of departure to move our knowledge forward. In the following subsections, the definition of the research questions, the search and screening process, the inclusion/exclusion criteria, and the data extraction are presented in depth.

2.1. Research questions

As recommended by Okoli and Schabram (2010), a review must define clear and concise research questions. Thus, two questions were defined for the current review and subsequently served as a guide: (1) What are the building blocks of digital platform ecosystem governance? And (2) What do we need more research on? With the first question, the review aims to identify what digital platform ecosystem governance entails by grouping and connecting the concepts identified in the extant research. With regard to the second question, this review aims to identify areas for future research. Both questions are in line with the general purpose described in the introduction of the Method section. As for the audience (Okoli & Schabram, 2010), this review is mainly beneficial to researchers—both experienced and inexperienced—in the topic at hand. The review can also be helpful for practitioners who want to govern their ecosystems in a structured manner.

2.2. The search and screening process

The search process was organized according to the guidelines by Webster and Watson (2002) and Okoli and Schabram (2010). To obtain a multidisciplinary overview of platform governance, this review searched for articles published in the fields of IS, strategic management, innovation, organization studies, and economics. The goal was to identify articles of validated quality. As major contributions are likely to be published in leading journals (Webster & Watson, 2002), this review searched only within journals in the Association of Business Schools' Academic Journal Quality Guide (updated in 2021)[1] ranked ABS 3 or higher. The database Scopus was used to perform this search.

This review used the following keywords: platform OR ecosystem OR "two-sided market" OR "two sided market" OR "multi-sided market" OR "multi sided market" AND governance OR governing OR orchestrat* OR control. These keywords were chosen to account for a breadth of fields and differing terminologies for digital platform ecosystems and governance. In fact, in the field of economics, the term "two-sided or multi-sided market" is often used when referring to platform ecosystems; whereas in the management and IS fields, "platform ecosystems" is more common. Moreover, "governance" is also described by using different terms as orchestration or control.

The search was performed on the articles' keywords because it was expected that relevant articles would use these words in their keyword sections. In Scopus, two ABS journals (e.g., the Journal of the Association for Information Science and Technology (JASIST), within the IS field, and the American Economic Review: Insights, within economics) were not available. To include them, a search was performed on these journals' websites. As for the former, the same query used in Scopus was used to search within the keyword section. Regarding the latter, as the search within keywords was not possible, the search was performed by going through all the articles, which was also possible due to a manageable number of issues.

Next, the search was extended to four of the most prestigious international IS conferences, as recommended by Webster and Watson (2002). This choice was also based on the fact that the conferences' review process can be considered similar to that of journals. In addition, conferences present shorter reviews and publication times, which allows for a more comprehensive overview of the research status. To search for conference papers, the same query used in Scopus was used in the e-library of the Association for Information Systems (AIS) within the abstract field because this database does not allow searching within keywords. The selected conferences were the International Conference on Information Systems (ICIS), the European Conference on Information Systems (ECIS), the Hawaii International Conference on System Sciences (HICSS), and the Americas Conference on Information Systems (AMCIS). In total, this paper searched 135 journals, along with the published proceedings for four international conferences.

This paper set the temporal limitations to contributions from January 2010 to mid-July 2023 because digital platform ecosystem governance is a relatively new topic that only became of interest to researchers during this period (Mukhopadhyay & Bouwman, 2019). The search resulted in 311 journal articles and 372 conference papers, which were screened first by title, then by abstract, and finally by the articles themselves. This screening process resulted in 37 journal articles and 40 conference papers.

Then, a backward and forward search was performed (Webster & Watson, 2002). In particular, a backward search was performed by reviewing the citations for the identified articles through the main search. This process resulted in an additional 18 papers, 16 journal articles, and 2 conference papers. The forward search aimed to identify articles citing the articles identified in the previous two steps (main search and backward search). This step was done on Scopus, which allowed for the tracking of the citations with the function "view cited by." For articles not available in Scopus, the forward search was performed on Google Scholar by screening among "cited by" articles. In this step, 8 additional papers (7 journal articles and 1 conference paper) were identified. For these steps, again, only articles ranked ABS 3 or higher in the selected fields of research during the chosen timeframe were considered. Ultimately, 103 papers (60 journal articles and 43 conference papers) were included in this review. The overall search and screening process is summarized in Fig. 1.

[1] In the ABS list, the field of information systems (IS) is referred to as "Information Management."

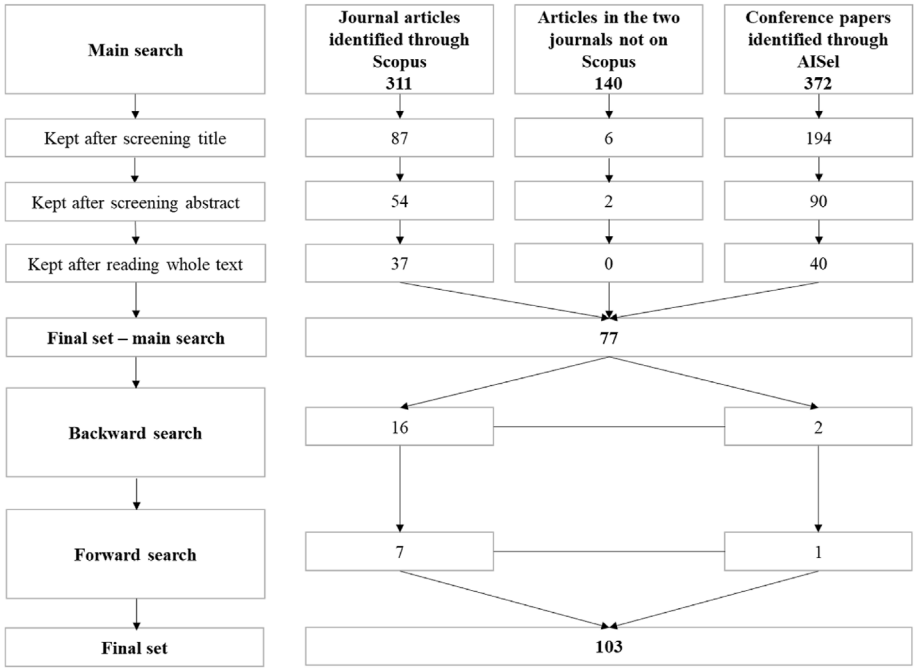

Fig. 1. Search and screening process.

2.3. Inclusion and exclusion criteria

As stated, this review targeted journal articles and conference papers of validated quality in the timeframe from January 2010 to mid-July 2023. Only articles written in English were included. For conference papers, only completed research papers were included. The search included articles on the following topics: digital platform (ecosystem) governance, governance mechanisms, components, dimensions, contingency factors, and outcomes. Articles that touched on these topics were also included. However, only those that addressed the digital platform ecosystems of private companies were considered. Private and public organizations differ in several factors, such as the legal and political environments in which they operate, the strength of their competitive pressures, and their internal structures and processes (Rainey et al., 1976). For example, public companies are less autonomous in their decisions, purposes, and operations; are usually interested in collective purposes; and must meet the fairness and integrity expected by citizens. In comparison, private companies usually have individual goals and aim at efficiency. It is also reasonable that the actors involved would be different, such as complementors and/or users in private companies' platforms and hospitals, government agencies, or other institutions in public platforms. In line with this focus, articles on the following topics were excluded: public sector governance, governance/management of natural resources (e.g., forests, coasts, and wetlands), policymaking, welfare, and education. Aside from the topic, these articles focused on the country level, far from the ecosystem and governance at the heart of this review. Moreover, articles about the governance of specific elements within the ecosystems (e.g., IT or data governance) and those that did not match the inclusion criteria were excluded.

2.4. Data extraction

From each article, the following information was extracted: the full reference, name of the journal or conference, and year; research method(s); theoretical frameworks and reference theories used; main topic; context and case if available (e.g., software-based platform, blockchain platform, crowdwork, and Apple and Amazon); level of analysis; research questions; contingency factors and summary of the study; and gaps for future research. The data extraction was performed on an Excel spreadsheet, where each extracted information was written in a different column.

3. Platform ecosystem governance and model

To synthesize and consolidate the extant research, the findings of this review are structured according to a conceptual model (Miles & Huberman, 1994). Conceptual models explain, in a graphical or narrative way, the main aspects to be studied (e.g., factors, variables, and constructs) and their presumed relationships, as identified from theories, experiences, and the general objectives of a study (Miles & Huberman, 1994).

The creation of the conceptual model in the current study was based on an analysis of the selected set of articles. Extracting information from the different articles was key to identifying the main components of digital platform ecosystem governance to be investigated and their relations. As specified in the previous section (Section 2.4), this extraction was performed by filling in an Excel spreadsheet with multiple columns for different information. This approach facilitated filtering of the collected data and grouping papers in a concept-centric fashion and along their level of analysis (i.e., focusing on the same aspects of governance, such as control or openness; Webster & Watson, 2002). Then, concepts were further aggregated into the components to be studied in this review (e.g., governance mechanisms). The aggregation was based on the researcher's understanding of the literature and the aims of this review. This process was inspired by previous studies on governance in different contexts (e.g., IT governance or corporate governance; Tiwana et al., 2013; Weill and Ross, 2004) and the context-mechanism-outcome scheme (Pawson et al., 1997). The resulting conceptual model is shown in Fig. 2.

The conceptual model has five components. **Governance structure** refers to who owns and chooses the digital platform ecosystem's use, development, and architecture. **Governance mechanisms** refer to the means used to implement the chosen governance structure. **Governance scope** refers to the expansiveness of the governance mechanisms, such as on what relationship (e.g., on all complementors or some of them) and where (e.g., on or off platform) they act. These three components are the core building blocks of digital platform ecosystem governance and are organized in a funnel wherein the components are narrowed down. **Contingency factors** refer to the elements that affect the core blocks or

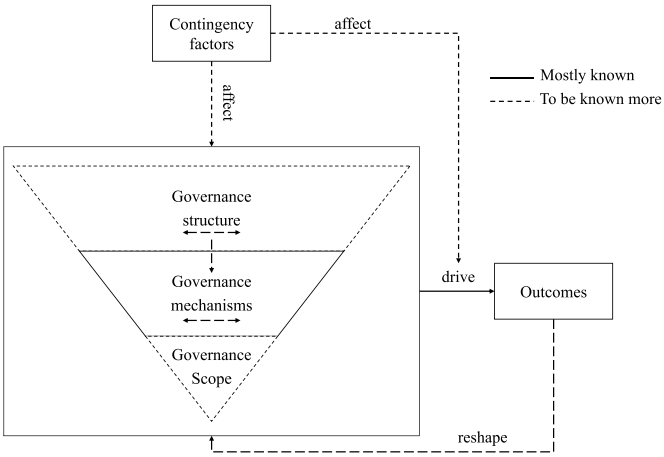

Fig. 2. Conceptual model for platform ecosystem governance.

moderate the outcomes of digital platform ecosystem governance. These factors include the characteristics of the actors involved, as well as the characteristics of the platform and of the market in which the digital platform ecosystem operates. **Outcomes** refer to the results achieved by digital platform ecosystem governance that mainly relate to value dynamics, that is, value co-creation and value capture.

4. Findings

In the following subsections, this paper presents and discusses the current state of the literature using a detailed version of the conceptual model above (Fig. 2). This detail version is shown in Fig. 3.

4.1. Governance structure

Governance structure includes ownership, decision rights, and architecture. **Ownership** refers to who owns a platform, whether it is a single actor (proprietary ownership), a multitude of actors (shared ownership), or a peer-to-peer network (decentralized ownership; Perscheid et al., 2020; Schultze et al., 2021; Tiwana et al., 2010). Ownership is also connected to the definitions of roles comprising the distribution of power and the number of sides within a platform ecosystem (Schreieck et al., 2016). Common roles in a platform ecosystem are the platform owner (an organization that offers the technical platform and sets standards and assets on which other organizations can build), complementors (other organizations that use the platform's technology to create complementary products and/or services; refers mainly to the supply side), and the users of those products and services (demand side; de Lima Fontão et al., 2019; Heimburg & Wiesche, 2022). Some researchers have

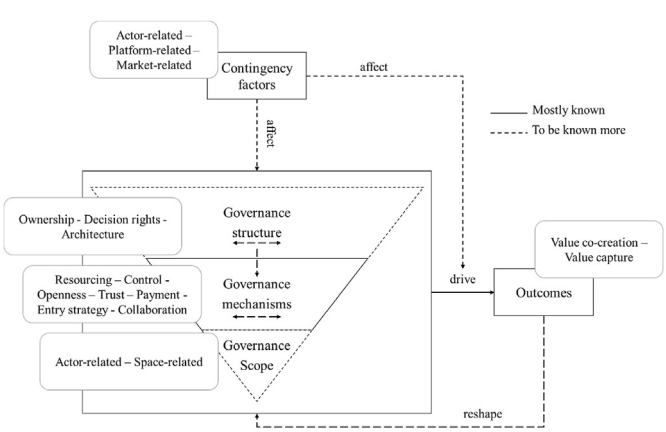

Fig. 3. A detailed conceptual model of platform ecosystem governance.

pointed out that these roles are unstable and that actors can also occupy dual roles, such as platform owners and complementors or complementors and users (Burda et al., 2022; Heimburg & Wiesche, 2022). Foss et al. (2023) suggested that a company becomes a platform owner not only when building an ecosystem in the first place, but also by virtue of successfully sustaining the ecosystem.

Decision rights indicate who has the authority and responsibility to make specific decisions, such as those related to the features, functionality, design, and implementation procedures of complementary products (Tiwana et al., 2010). Decision rights can be divided between platform owners and complementors, and they balance complementors' autonomy and coordination within the ecosystem (Tiwana et al., 2010). Decision rights can affect complementors' participation within a platform, because they may increase or decrease costs and risks incurred when complying with specifications (Dellermann & Reck, 2017). Platform owners who have decision-making authority can exercise this right to their benefit, but diverse interests must be balanced if the platform owners want to attract and retain complementors (Schulze et al., 2021; Tiwana et al., 2010). Focusing on ownership and decision rights, researchers have distinguished between centralized, decentralized, and autonomous digital platform ecosystems (Perscheid et al., 2020; Riasanow et al., 2018; Schulze et al., 2021; Werner et al., 2020). In centralized arrangements, decision rights are mainly in the hands of the platform owner, especially with regard to whether a decision is to be implemented and how its performance is measured. In decentralized platform ecosystems, decision rights are handled by the community (i.e., the users of the platform). In autonomous platform ecosystems, such as blockchain-based ecosystems, decision rights are encoded in smart contracts. Schulze et al. (2021) identified different archetypes of digital labor platforms that go beyond the centralized and decentralized dichotomy by showing that decision rights can be distributed differently during the stages of a service transaction.

Architecture refers to the ecosystem's blueprint, which describes the parts within the ecosystem and how they are related (Tiwana et al., 2010). The architecture comprises the central platform, the interfaces that mediate between the platform and its complements, and the complements. In platform ecosystems, authority also comes from the architecture, specifically from the control of the architecture at the center of the ecosystem (Kretschmer et al., 2022). Several researchers have stated that the interrelation between the division of decision rights and platform architecture affects complementors' engagement and a platform's success (Jacobides et al., 2018; Saadatmand et al., 2019; Tiwana et al., 2010). For example, Tiwana et al. (2010) suggested that a modular architecture (i.e., an architecture wherein the core and extension modules are loosely coupled so that complementors can act more independently) reinforces the benefits of decentralizing decision rights by reducing the costs incurred by complementors, thus accelerating the platform's evolution. According to Saadatmand et al. (2019), complementors' engagement is low when decision rights over architecture are allocated to a single actor, whereas engagement is high when decision rights are distributed among ecosystem complementors that have the same opportunities for value capture. Some researchers have suggested that architecture and its changes should also be dynamically related to governance mechanisms (Jovanovic et al., 2022b; Mei et al., 2022), which will be described in the following subsection (4.2).

4.2. Governance mechanisms

The literature has identified several governance mechanisms that are often based on boundary resources. Thus, before each mechanism is described, boundary resources should first be clarified. Boundary resources are technical and social tools that enable interactions between the platform owner and complementors (Ghazawneh & Henfridsson, 2013). Whereas technical boundary resources make the development of complementary products or services more feasible and improve platform access, social boundary resources transfer knowledge to ensure better

understanding and interaction among actors (Ghazawneh & Henfridsson, 2013; Luo et al., 2019; Miric et al., 2023). Examples of technical boundary resources include application programming interfaces (APIs) and software development kits (SDKs); examples of social boundary resources include guidelines and knowledge that support complementors (Foerderer et al., 2019; Ghazawneh & Henfridsson, 2013; Luo et al., 2019). These resources can be standardized for all complementors (e.g., general agreements) or individualized to account for specific needs (e.g., a personal contact complementors can refer to; Engert et al., 2022).

As previously mentioned, several governance mechanisms exist, such as resourcing, control, openness, trust, payment, entry strategy, and collaboration. In what follows, each governance mechanism will be presented in detail.

Resourcing aims to stimulate complementors toward creating increasingly diverse products or services (Ghazawneh & Henfridsson, 2013; Halckenhaeusser et al., 2020). Providing boundary resources is not a one-way mechanism from the platform owner to the complementors. Rather, boundary resources can be developed through interactions and fine-tuning between these actors (Eaton et al., 2015; Foerderer et al., 2019; Mohagheghzadeh & Svahn, 2016a). For example, if complementors perceive that certain boundary resources have limited functionalities, they may decide not to use them (Mohagheghzadeh & Svahn, 2016b), develop new boundary resources (i.e., self-resourcing; Ghazawneh & Henfridsson, 2013), or require additional boundary resources be provided to achieve their goals (Glaser et al., 2020; Rickmann et al., 2014). In the process of building a platform ecosystem, involving lead complementors (e.g., those whose boundary resource requirements are representatives of a broader group because they face such requirements earlier than others owing to their technical expertise and use of boundary resources) in designing boundary resources can result in boundary resources that are both attractive for complementors and aligned with the interests of incumbent platform owners (Weiss et al., 2023).

The mechanism of **control** refers to maintaining control over a platform ecosystem and attempting to align complementors with the platform's purpose (Ouchi (1980) in Goldbach & Benlian, 2015b). Given that principal–agent relationships do not hold in ecosystems, control acts more to coordinate actors than to mitigate agency risks, that is, when the agent acts for their benefit and against the principal (Constantinides et al., 2018; Tiwana, 2015). Different types of control exist. Control can be *informal* when fostering culture, common values, shared beliefs, and norms to influence complementors' behavior (so-called *clan control and self-control*), or *formal* when complementors' outputs, processes, tasks, and inputs must comply with specific criteria (Askay, 2017; Constantinides et al., 2018; Croitor et al., 2021; Lin et al., 2023; Schreieck et al., 2016; Tiwana, 2015; Tiwana et al., 2010). Input control has recently received increasing attention, and researchers are investigating how it is perceived by complementors (e.g., its fairness) and how it affects platforms (Croitor & Adam, 2020; Croitor & Werner, 2021; Song et al., 2018; Thies et al., 2018), sometimes also in combination with other types of control (e.g., self-control; Croitor et al., 2021). Due to the growth of gig platforms, the emergence of algorithmic control (Hirsch et al., 2023; Wiener et al., 2021), an automatic control that does not require human presence and is based on the use of algorithms and related digital technologies (e.g., smartphone apps; Hirsch et al., 2023), has also been noted. Its different forms, transparency, and resistance and tension from the workers' perspectives are among the main aspects of algorithmic control investigated by researchers (Chen et al., 2023; Göttel, 2021; Hirsch et al., 2023; Jiang, 2023; Möhlmann et al., 2021; Weber et al., 2022).

Control is a dynamic mechanism due to actors' interactions and the ways in which technological features are understood and used; thus, it changes over time between formal and informal control and can originate from the intentions of platform owners or other participants (Ens et al., 2023; Tiwana et al., 2010). For example, some researchers have pointed out that (quality) control can be more distributed and not only confined in the hands of the platform owner, and that centralized and distributed control types can complement each other (Tang et al., 2021).

Another mechanism is **openness**, which refers to who is allowed to join and use the platform and its resources (Ingram Bogusz & Kashyap, 2022). An open platform can stimulate experimentation and decrease innovation costs, whereas a closed platform allows for maintaining control and quality over complementors' contributions (Gawer & Cusumano, 2014; O'Mahony & Karp, 2022). Openness is often the result of other governance mechanisms. For instance, it can come from reduced input control (e.g., reducing the screening process for campaign submission to a crowdfunding platform) or from resourcing APIs that grant access to a platform's core module (Eaton et al., 2015; Ghazawneh & Henfridsson, 2013; Wessel et al., 2017). Determining an adequate level of openness and adjusting it over time is a challenging and strategic decision that must account for complementors' perceptions of openness and for its potential impacts on value co-creation and its intensity (Benlian et al., 2015; Gawer & Cusumano, 2014; Hilkert et al., 2011; Schreieck et al., 2016; Schreieck & Wiesche, 2017; Van Dyck et al., 2021; Wessel et al., 2017).

Trust is the governance mechanism that aims to create and maintain relationships and a favorable environment (Goldbach & Benlian, 2015a; Guo et al., 2021). Trust enhances actors' willingness to participate in an ecosystem and is a prerequisite for the ecosystem's success (Hurni & Huber, 2014; Riasanow et al., 2018; Zhang et al., 2020). Trust can be placed in the platform or in other actors. On the one hand, *trust in the platform* may come from different sources, such as technologies and institutions. For example, trust can be encoded in smart contracts on blockchain-based platforms, or it can come from terms and conditions that form the basis for dispute resolution (Guo et al., 2021; Perscheid et al., 2020). On the other hand, *trust in actors* can be shaped through online and offline interactions, conflict resolution mechanisms, and verification, ratings, and review systems, all of which can help reduce information asymmetry and exchange uncertainty, as well as increase transparency (Guo et al., 2021; Jovanovic et al., 2022a; Zhang et al., 2020). If actors trust each other, they are more likely to spread the platform's norms, values, and common goals, thus encouraging others to act according to them and be more committed (Goldbach & Benlian, 2015a; Guo et al., 2021). Trust can also be an outcome of other governance mechanisms, such as payments (see below) and informal and formal controls (Goldbach et al., 2014; Zhang et al., 2020).

The mechanism of **payment** refers to money flows and distributions, such as pricing, revenue sharing, and payment for the provision of services (Gol et al., 2019a; Schreieck et al., 2016). This governance mechanism is mainly used to aid network effects and overcome the chicken-and-egg problem in the early stages (building) of an ecosystem (e.g., Sterk et al., 2022). In the case of two-sided platforms, one of the two sides is often subsidized to attract a critical mass of actors on the other side. Payment can also be useful in later stages, when owners aim to sustain their ecosystem or capture value. In this case, payments can refer to service fees that platform owners can introduce to capture value when they do not have the capabilities to enter a complementary market (Zhu & Liu, 2018).

The governance mechanism of the **entry strategy** refers to the decision of platform owners to enter complementors' markets by releasing their own complements that overlap with complementors' products and services (Foerderer et al., 2018; Wen & Zhu, 2019). Entering complementors' markets is a way to control and stimulate them, as well as to appropriate value (Foerderer et al., 2018; Zhu & Liu, 2018). However, the impact of this governance mechanism seems ambiguous. Entry can increase consumer demand and, therefore, bring new ideas and opportunities for innovation, especially for larger and more diversified complementors (Foerderer et al., 2018). However, it may also stunt complementors' innovation due to increased competition, especially for those with popular products, and it can benefit consumers because of reduced shipping costs (Zhu & Liu, 2018). Other times, entry makes complementors defensive and prompts them to divert offline customers away from the platform, thus reducing demand and hurting the entire

platform (He et al., 2020). Researchers have also examined the effects of potential entry on complementors, showing that innovation may shift to less popular areas or may be reduced; meanwhile, prices may be increased (Wen & Zhu, 2019).

Aiming at reconciling these divergent results, a recent study (Shi et al., 2023, in press) showed that it is the early or late timing of a platform owner's entry that makes a difference: early timing indicates the owner's commitment to grow the market's prosperity and its focus on value co-creation, whereas later timing suggests platform owners' intention of capturing value.

Finally, **collaboration** is the mechanism used by platform owners to team up with complementors (Halckenhaeusser et al., 2020; Schreieck et al., 2016; Schreieck & Wietsche, 2017). Collaboration can take different forms. For instance, Rietveld et al. (2019) investigated the use of selective promotion (e.g., featuring specific apps on the home screen of the app store or creating special marketing campaigns) to manage the ecosystem's value. Selective promotion increases the success of individual complementary products, but it can also increase competition, which, in turn, affects other complementors' incentives and bargaining power. Cenamor and Frishammar (2021) invited platform owners and complementors to unite forces in the development and commercialization of complementary products to aim at higher performance in terms of unit sales. Rewards are another form of collaboration (Halckenhaeusser et al., 2020). For instance, through partnership ranks, complementors can be awarded greater resources and higher responsibilities (Wareham et al., 2014). Awards can also guide complementors' attention toward a market niche, even if they may multihome (e.g., join and operate their complements on other platforms; Foerderer et al., 2021).

Some researchers have suggested that platform owners can and should use both collaborative and competitive approaches (e.g., entry strategy) to foster complementors' engagement in the short and long run (Engert et al., 2023).

4.3. Governance scope

Governance scope refers to the expansiveness of governance mechanisms, that is, to what relationships (actors involved, e.g., all complementors or some of them) and where (on or off platforms) they act. In the first case, the scope is actor-related; in the second case, the scope is space-related.

For the **actor-related scope**, governance mechanisms can affect the relationship between the owner and all (or some) complementors and the relationships among the complementors (or users). Most studies have focused on the *relationship between the platform owner and complementors* (Foerderer et al., 2018; Ghazawaneh & Henfridsson, 2011; Goldbach et al., 2018; Heimburg & Wiesche, 2022; Mohagheghzadeh & Svahn, 2016a, 2016b). The complementors investigated were mostly external companies, with the only exception being the work of Glaser (2020), which examined internal complementors (i.e., teams that belong to the platform provider and that develop platform add-ons). The platform owner usually takes an ecosystem-wide approach, which means that it uses the same standardized rules (e.g., rights and duties), values (e.g., promoted in codes of conduct), and technologies (e.g., boundary resources) to govern the ecosystem. Research agrees that this approach saves costs, but its impact on value co-creation and capture seems ambiguous. For instance, some researchers found that a standardized approach leads to less novel but more commercially successful products (e.g., higher sales; Miric et al., 2023). Others found that it may hamper value co-creation opportunities, suggesting that a platform owner focuses on its *relationship with a set of complementors* and on their local needs through additional, individualized resourcing or partnership programs in which the partner level can be self-selected, or through the use of more flexible rules (Engert et al., 2022; Glaser, 2020; Huber et al., 2017; Hurni et al., 2021; Kindermann et al., 2022; Wareham et al., 2014). Platform owners must recognize that complementors are autonomous and active actors who will try to ensure that their specific needs are not lost, which is essential for value co-creation (Hurni et al., 2022).

Thus far, researchers have paid very little attention to how governance mechanisms can affect the *relationships among complementors (or users)*, with a couple of exceptions. For example, Zhang et al. (2022) investigated how openness can affect knowledge-sharing activities among complementors to balance value co-creation and capture. Burtch et al. (2019) focused on interactions among users through peer recognition (e.g., in the form of "likes" and awards) that can encourage more, but less novel, content generation.

Space-related scope refers to whether the focus of governance mechanisms is *online* (on the platform) and/or *offline* (outside the platform). Zhang et al. (2020) showed that online and offline practices interact and encourage varied interpersonal relationships in the context of sharing platforms. Wang and Nandhakumar (2017) suggested growing and governing a platform and its sides by focusing on online and offline spaces, such as, respectively, chat rooms, information standardization, or more APIs, and through user meetings, peripheral gifts (e.g., t-shirts) or trusteeship management. Although most articles focus on online space, focusing on off-platform activities can foster relationships, create a sense of community, and incentivize platform participation (Suvivuo et al., 2023; Wang & Nandhakumar, 2017). Sometimes, however, these off-platform events may discourage platform users from connecting on the platform (Zhang et al., 2020).

Thus far, this review has provided a description of the three core building blocks of digital platform ecosystem governance. In the following two subsections, the focus will be on contingency factors and outcomes.

4.4. Contingency factors

Contingency factors refer to elements that affect the core building blocks or mediate the outcomes of digital platform ecosystem governance. There are three types of contingency factors: actor-, platform-, and market-related.

Actor-related contingency factors refer to the *individual characteristics of platform owners and/or complementors*. For platform owners, whether they are established companies or start-ups makes a difference, for instance, when it comes to the extent of openness or in relation to their bargaining power in building or sustaining platforms (Floetgen et al., 2023; Schreieck & Wiesche, 2017; Sterk et al., 2022; Wang & Nandhakumar, 2017). A platform owner's bargaining power, previous decisions, and capabilities (which can also be learned) can affect how governance is enacted and changed from emergent to mature stages (e.g., whether the governance scope will address the entire ecosystem or local needs), or the possibilities of co-creating and capturing value through the platform (Foss et al., 2023; Helfat & Raubitschek, 2018; Huber et al., 2017; Kindermann et al., 2022; Schreieck et al., 2021; Uzunca et al., 2022).

For complementors, some researchers have shown that their size (in terms of app portfolios), diversification (in terms of targeted categories), popularity, and capabilities affect their perceptions of and responses to entry strategies (Foerderer et al., 2018; Wen & Zhu, 2019; Zhu & Liu, 2018). Other researchers have mentioned that complementors' complex or routinized knowledge, locations, perceptions, and assessments, or their products' reputation affect their activities and the effectiveness of governance mechanisms (Croitor et al., 2021; Croitor & Werner, 2021; Goldbach et al., 2018; Gutt et al., 2019; Koo & Eesley, 2021; Wiener et al., 2021; Zhang et al., 2022).

Platform-related contingency factors refer to a *platform's characteristics*, such as its evolutionary stage and type. Van Dyck et al. (2021) reported that emergent ecosystems need to strategically balance resource openness. Schreieck et al. (2021) suggested that creating value is more important than capturing value when building ecosystems, yet value capture cannot be neglected in the early stages or it will be more difficult afterward. For the platform type, crowdfunding platforms may need to quickly increase the critical mass of funders due to the short time horizon

of their campaigns (i.e., days or weeks) compared with other types of platforms based on games or apps that are available for a longer time (i.e., months or years; Thies et al., 2018). Another example is blockchain-based platforms, which are usually more open than other types of platforms that require users and complementors to authenticate (Pereira et al., 2019). Another platform-related contingency is platform strategy, which, if generalist (specialist), increases (decreases) the range of possibilities for reducing conflicts between platform owners and complementors, thus increasing (decreasing) outcomes (Islam et al., 2023).

Market-related contingency factors are *characteristics of the market* in which the platform ecosystem operates. A competitive market can reduce complementors' investments in a platform (Wang, 2021). In the case of crowded markets, platform owners can reduce openness to lower competitiveness and foster complementors' cooperation (Zhang et al., 2022). Markets with high demand growth increase opportunities for collaboration toward value co-creation, whereas markets with low demand growth (more mature markets) have poor opportunities for value co-creation and make actors compete to capture value (Uzunca et al., 2022). Another aspect may be the higher regulated environment that prompts platform owners to (re)shape platform functionalities and decide on the level of openness (Ingram Bogusz & Kashyap, 2022). Another study concluded that legitimacy (i.e., what is considered desirable, proper, or appropriate within a context) can affect a platform's openness and sustainability and even has a heavier weight along a platform's evolution (Brandwijk & de Reuver, 2023).

4.5. Outcomes

Outcomes are the results expected or achieved by the chosen governance structure, mechanisms, and scope and are mediated by contingency factors. Outcomes essentially refer to a platform ecosystem's value dynamics—that is, value co-creation and value capture.

Value co-creation is generally understood as *innovations* (development activities) or *transactions* between the sides of a platform ecosystem (Foerderer et al., 2018; Foss et al., 2023; Gawer, 2014; Gawer & Cusumano, 2014; Gol et al., 2019b; Goldbach & Benlian, 2015b; Huber et al., 2017; Inoue, 2021; Zhang et al., 2020). In digital platform ecosystems, value is co-created because it is shaped through interactions by platform owners and complementors or among complementors (Cenamor & Frishammar, 2021; de Vasconcelos Gomes et al., 2021; Halckenhaeusser et al., 2020; Schreieck & Wiesche, 2017; Selander et al., 2013; Zhang et al., 2022).

Value co-creation is often described and measured in different ways, such as through product performance, increased quality of app notifications, user satisfaction, or the tangible and intangible benefits resulting from combining partners' resources with the soon-to-be realized value that can come from acquiring important clients or serving important clients' needs (Cenamor & Frishammar, 2021; Cennamo & Santaló, 2019; Claussen et al., 2013; Huber et al., 2017). Therefore, value is dependent on context and can be affected by the governance structure, governance mechanisms, and/or governance scope. For instance, for the structure, the configuration of the decision rights and architecture can affect complementors' engagement, and the centralization of decision rights can affect their uncertainties and hazard costs, thus influencing their willingness to invest in the development of complementary products (Dellermann & Reck, 2017; Niedermayer, 2013; Saadatmand et al., 2019). For mechanisms, higher value co-creation can be achieved through collaboration between owners and complementors, whereas formal control (e.g., rigid regulations) can stunt it (Cenamor & Frishammar, 2021; Eaton et al., 2015; Schreieck et al., 2016). For governance scope, Zhang et al. (2022) suggested that fostering knowledge sharing among complementors can generate value, whereas Huber et al. (2017) suggested that addressing local needs can lead to higher value.

Co-creating value is not a straightforward process. One of the reasons is its interaction with **value capture**, that is, the appropriation of part of the co-created value—a topic that has received less attention (Foss et al., 2023; Schreieck et al., 2021). The dynamics of value co-creation and value capture must be well balanced because if a platform owner captures a large portion of the co-created value or distributes it in a way that seems unfair to complementors, the recruitment of potential complementors can be difficult (Jacobides et al., 2018; Sterk et al., 2022). Another reason is related to mediation by contingency factors. For instance, the outcomes of platform ecosystem governance may be uncertain because of changes in the way complementors, each with varying degrees of product popularities, adjust their value co-creation and capture strategies in response to governance mechanisms (Wen & Zhu, 2019).

5. Discussion and research agenda

As the findings and the conceptual model demonstrate, platform ecosystem governance is complex and multifaceted; thus, it cannot be adequately understood when looking narrowly at some of its building blocks. Platform ecosystem governance occurs within a context and is thus influenced by contingency factors related to the actors, platforms, and markets in which they operate. The core building blocks of platform ecosystem governance (governance structure, governance mechanisms, and governance scope) or their elements can also serve as contingency factors for one another. For instance, within a governance structure, the definition of ownership affects the division of decision rights (Hein et al., 2020). When a platform ecosystem has a single owner, this actor will allocate to themselves the full (or at least the largest) share of decision rights; in other cases, decision rights can be divided among actors, as in the hybrid and collective modes of governance in the work of O'Mahony and Karp (2022).

Ownership also affects an ecosystem's evolution by affecting outcomes (directly or indirectly via governance mechanisms) and changing how governance mechanisms are implemented (Gol et al., 2019a; Hein et al., 2016; Tiwana, 2013; Tiwana et al., 2010). For example, ownership—and the bargaining power that comes from it—has been described as a commitment device that can influence complementors' willingness to invest in complementary product development (Niedermayer, 2013) and hence, their participation in value co-creation. How ownership affects the implementation of governance mechanisms can be understood using the following examples: Facebook, which has a centralized platform ecosystem, is the only organization that establishes and changes the platform's governance mechanisms, whereas Uber, which is midway between a centralized and a decentralized platform ecosystem, has full control over prices but gives users quality control through a driver rating system (Hein et al., 2016).

Governance mechanisms can be implemented to orchestrate an ecosystem widely or more locally and to coordinate complementors or their interactions, both online and offline, thus affecting the governance scope. Sometimes, governance scope may also affect governance mechanisms, such as when physical relationships among actors lead to higher levels of trust among them.

The core building blocks of governance structure, governance mechanisms, and governance scope, together with contingency factors, affect the generation of outcomes in terms of value co-creation and value capture. In turn, these outcomes can reshape governance mechanisms and governance scope, which can be investigated further (Huber et al., 2017; Wessel et al., 2015). In addition, deepening our understanding of the relationships among and within the building blocks is key to advancing our theoretical and practical knowledge. Thus, avenues for future research are presented in the following subsections.

5.1. Research agenda

Based on what researchers have investigated thus far, along with the changes in the platform ecosystem scenario and the gaps identified by extracting information from the gathered articles, this review suggests a research agenda for each building block of the proposed conceptual

model (Fig. 2). These avenues are considered important for enriching our knowledge and understanding of platform ecosystem governance. Considering the strong connections among the building blocks, research on each block is related to other blocks as well. Table 1 provides a summary of the identified avenues, specifying the research areas (building blocks), the topics within them, and the related research questions.

5.1.1. Future research on governance structure

As described in 4.1, governance structure comprises ownership, decision rights, and architecture. The interplay between ownership and decision rights gives rise to platform ecosystems with different degrees of (de)centralization. Researchers have mainly focused on centralized platform ecosystems and their focal actors, with very little attention paid to a more distributed model of organization (de Reuver et al., 2018; Saadatmand et al., 2019; Selander et al., 2013). Some exceptions are those works related to blockchain-based ecosystems (Burda et al., 2022; Pereira et al., 2019; Perscheid et al., 2020) and the work of O'Mahony and Karp (2022), which focused on collective platform governance in which those affected by the rules can participate in their making. Therefore, first, future research may investigate (more) decentralized forms of platform ecosystems. Within these decentralized platform ecosystems, collaborative platform ecosystems appear interesting for future investigation. In collaborative platform ecosystems, the governance structure (especially ownership and decision rights) is shared among multiple actors, none of which is the owner of the ecosystem, and may also be organized in some sort of consortia (Hein et al., 2020). Collaborative platform ecosystems are a new type of ecosystem that emerges around goals that are best approached in collaboration rather than in isolation. For instance, some ecosystems have been launched in the aquaculture industry in Norway to address sustainability challenges, while others have been launched in other industries, such as healthcare and oil and gas. Therefore, focusing on such ecosystems would allow us to extend our knowledge apart from centralized platform ecosystems. Doing so will also allow us to better grasp the evolving business scenario, which, aside from theoretical development, will provide more relevant guidelines for practitioners. Moreover, depending on the collaborative ecosystem's goal(s), future research may add knowledge of how digital platform ecosystems relate to issues of societal and global interest (de Reuver et al., 2018).

Different aspects could be considered when investigating such collaborative platform ecosystems. For example, it could be worth exploring how decision rights are allocated. One possible question is, "Based on what criteria (e.g., based on size or power) would roles and decision rights be allocated in a collaborative platform ecosystem?" Researchers may also focus on how value is co-created, distributed, and captured in such settings and on how value, in the first place, can be defined at the firm and ecosystem levels.

Second, researchers could investigate how governance structure affects the choice of governance mechanisms. The extant literature has been silent on this point despite the recent work of Floetgen et al. (2023) suggesting that platform ecosystems led by a single actor (incumbent or start-up) and those led by a strategic alliance (which may be seen as a decentralized ecosystem) implement different governance mechanisms. Therefore, future research may follow a multiple case study or undertake a survey across several digital platform ecosystems to explore whether and how platform ecosystems led by a single actor or by a consortium choose and employ their governance mechanisms. To address this research direction, future studies may also conduct single-case studies of collaborative platform ecosystems and investigate questions such as "Does a platform owner's entry into complementary markets make sense in a consortium-led platform ecosystem? If so, which of the companies leading a platform ecosystem, and on which basis, enter complementary markets, and how is value divided?"

Conducting research along these lines will provide us with a more in-depth understanding about whether collaborative (or more

Table 1
Research agenda.

Research area (building block)	Topic(s) of interest	Research questions
Governance structure	Collaborative platform ecosystems	Based on what criteria (e.g., based on size or power) would roles and decision rights be allocated in a collaborative platform ecosystem? How is value co-created and appropriated in collaborative platform ecosystems? Do collaborative platform ecosystems present challenges other than those of centralized platform ecosystems?
	The impact of governance structure on governance mechanisms	How does the governance structure affect the choice of governance mechanisms? Does a platform owner's entry into complementary markets make sense in a consortium-led platform ecosystem? If so, which of the companies leading a platform ecosystem and, on which basis, enter complementary markets? In this case, how is value divided?
Governance mechanisms	Strategic role and design of boundary resources	What makes boundary resources appealing for complementors? How should boundary resources be strategically designed? How can technical and social boundary resources be best combined? Do they have different impacts on complementors, or can these resources work together to stimulate complementors?
	A contingency model of governance mechanisms, contingency factors, and outcomes (comparing and combining them in different conditions)	Under what condition(s) is one governance mechanism preferred to another? How can governance mechanisms be best combined? How do governance mechanisms change across platform ecosystems and throughout the platform's life cycle?
Governance scope	Interactions among complementors Offline scope	What are the effects of (online and offline) governance mechanisms on complementors' interactions and their contributions toward collective knowledge and value co-creation?
Contingency factors	Relative importance of contingency factors, their interrelations, and their casual chains (if any)	What contingency factors matter most when building a digital platform ecosystem? What contingency factors matter most when sustaining a digital platform ecosystem?
	Governance mechanisms as contingency factors of platform participation (signaling role)	How do prospective complementors' perceptions of governance mechanisms affect their platform participation choices?
Outcomes	Outcomes at the platform level	What are the effects of governance mechanisms at the platform level?

(*continued on next page*)

Table 1 (*continued*)

Research area (building block)	Topic(s) of interest	Research questions
		Do governance mechanisms contribute to a platform ecosystem's health, integrity, and sustainability? What about the effects of governance mechanisms on the other side of a platform?
	Effects of outcomes on past governance choices	How do outcomes affect and reshape governance decisions?
	Relationship between value co-creation and value capture	How should the co-created value be best divided among actors?
		How does the value captured by a platform owner affect the stability of the entire platform ecosystem?

decentralized) platform ecosystems present governance challenges other than centralized ones, as well as how governance structure affects other governance choices down the funnel and the platform's future evolution (Burda et al., 2022).

5.1.2. Future research on governance mechanisms

Despite the large amount of attention paid to governance mechanisms, this building block requires additional investigation. First, given that many governance mechanisms are based on boundary resources, the strategic design and role of boundary resources can be explored further (Ghazawneh & Henfridsson, 2013). Not all boundary resources are mandatory to use; thus, future research may ask what makes them appealing for the development of complementary products and how platform owners should strategically design them. Mohagheghzadeh and Svahn (2016b) reported that intuitive boundary resources are an effective way to transfer design capabilities, in that they would cut down time for exploration. Weiss et al. (2023) suggested that incumbent platform owners could engage and collaborate with lead complementors in the design of these resources rather than push them into a sort of 'resistance and accommodation' (Eaton et al., 2015, p. 220). Aside from these works, however, our knowledge is limited. There are other questions that require more examination, such as "How can technical and social boundary resources best be combined?" and "Do they have a different impact on complementors, or can these resources work together to stimulate complementors?" Replying to these questions will help platform owners make more considerate choices related to their boundary resources in ways that facilitate innovation and interactions among actors.

Second, we still know little about how to choose specific governance mechanisms over others. Some researchers have compared formal and informal control and within these types, but often in simulated platform settings or have accounted for intentions rather than actual behaviors (Croitor et al., 2021; Goldbach & Benlian, 2015b; Goldbach et al., 2014, 2018; Goldbach & Kemper, 2014; Göttel, 2021). Therefore, comparing across governance mechanisms and in more real settings would enrich our understanding of the effectiveness of governance mechanisms. For instance, future researchers may compare control and resourcing or investigate how competition mechanisms (e.g., entry strategy) can be combined with other mechanisms.

Moreover, investigating the same mechanism in different types of digital platform ecosystems would also be beneficial. For example, Foerderer et al. (2018) suggested that entry may have different effects on complementors over time and may also depend on the platform ecosystem type, such as whether it is a consumer or enterprise platform. The choice of governance mechanisms along the evolution of a platform ecosystem also requires further research. There is strong agreement on using payment in the early stages; however, this cannot be taken for granted. For instance, Wang and Nandhakumar (2017) claimed that in launching platform ecosystems and growing their critical mass, startups can start with the side that is most approachable with existing resources, instead of the one that is more price sensitive.

Overall, there is a need to look at mechanisms and contingency factors (see also 5.1.4) in a more structured and comprehensive way. In this direction, a contingency model may be developed based on a quantitative study of how contingencies affect governance mechanisms and outcomes. Such a contingency model could be valuable for gaining insights into what comprises the best fit for a specific context, such as across platforms and along the platform life cycle. This knowledge would allow practitioners to make their governance decisions on a sounder basis and increase their effectiveness.

5.1.3. Future research on governance scope

Our current knowledge of how governance mechanisms shape interactions among complementors remains scant (Zhang et al., 2022). This is unfortunate because, even if interactions among complementors may not be directly associated with value co-creation (Kretschmer & Claussen, 2016; Rietveld et al., 2019), they are still relevant and can drive novel knowledge recombination and the creation of reusable knowledge and innovation in ways that can help other complementors achieve platform value co-creation (Adner & Kapoor, 2010; Wareham et al., 2014; Zhang et al., 2022). Moreover, shaping interactions among complementors toward a goal may contribute to improved management of the tension between individual versus collective identification described by Wareham et al. (2014).

Therefore, a first avenue for future research can aim at extending our knowledge of how governance mechanisms can foster complementors' interactions and their contributions to other complementors toward achieving value co-creation. Given the scant research on this theme, future studies may follow a qualitative approach to investigate the possible effects of governance mechanisms on complementors' interactions. Second, along this line of thought, future research may also investigate how adopting an offline scope (i.e., applying governance mechanisms outside the platform) may foster complementor interactions, thereby intersecting the two least investigated aspects of governance scope.

5.1.4. Future research on contingency factors

Platform owners' governance decisions must be tailored to the specific environment and needs. As shown in 4.4, the extant research has brought to the table several contingency factors at different levels. In 5.1.2, the theoretical and practical necessity of developing a contingency model that can relate the context, governance mechanisms, and outcomes has been stated. In this subsection, the first avenue for future research involves exploring the relative importance of contingency factors, their interrelations, and their casual chains (if any). Studying and evaluating these aspects may strengthen the contingency model and increase our knowledge of how to govern platform ecosystems in specific contexts throughout their evolutionary stages.

Second, future research may focus on the signaling role of governance mechanisms, i.e., how they are perceived by complementors and can act as contingency factors that affect complementors' consideration to join and participate in platform ecosystems (Adam et al., 2023; Kretschmer et al., 2022). Along the line of Adam et al. (2023), who find that a moderate degree of perceived input control maximizes prospective complementors' intention to join, future research may ask: "How does prospective complementors' perceptions of governance mechanisms affect their platform participation choice?" Answering this question may increase our understanding of how platform owners could best attract complementors, which in turn, may be useful when building and eventually sustaining an ecosystem. Moreover, addressing such a question can provide further insights into platform competition (Rietveld & Schilling, 2021).

5.1.5. Future research on outcomes

Thus far, the extant research has looked at outcomes in a narrow way. This is because the literature has generally looked at complementor-related outcomes, such as the innovations they have implemented or their stickiness to a platform (e.g., Ghazawneh & Henfridsson, 2010; Goldbach & Kemper, 2014; Göttel, 2021). However, platform ecosystem governance can do much more than this. For years, it has been recognized that platform ecosystem governance affects a platform ecosystem's evolutionary dynamics (Tiwana et al., 2010). However, our understanding of this impact is limited and untested (Mukhopadhyay & Bouwman, 2019; Song et al., 2018). Therefore, in enriching our knowledge along this line, researchers may first focus on the following questions: "Do governance mechanisms contribute to a platform ecosystem's health, integrity, and sustainability (Iansiti & Levien, 2004)?" "What about the effects of governance mechanisms on the other side of a platform?" Only a few works seem to have followed this direction. For example, Wessel et al. (2015, 2017) stated that abolishing input control, despite increasing the number of complementary products, reduces platform appeal. Song et al. (2018) and Thies et al. (2018) reported that control generates asymmetric cross-side network effects, thus undermining the idea of mutual cross-network effects between the two sides of platform ecosystems and self-sustaining growth. To increase our understanding of this research avenue, adopting a platform level of analysis may be key to accounting for the broad outcomes that platform ecosystem governance can generate. Adopting a longitudinal perspective is also fundamental in that the governance mechanisms may take time to generate outcomes and may affect the platform ecosystem's sides at different points in time (Song et al., 2018).

Second, we have limited knowledge about how outcomes affect and reshape platform ecosystem governance decisions made until that moment (e.g., governance mechanisms or scope). The abovementioned work by Wessel et al. (2015) provides some insights. In particular, the authors pointed out that measures for establishing a shared vision across all ecosystem participants (a sort of clan or informal control) were introduced to counteract the platform's reduced appeal brought about by a relaxation of formal control. Wang (2021) found that controlling complementors can benefit consumers and platform owners. Huber et al. (2017) suggested further investigating how a governance scope that targets a set of complementors and their local needs impacts other complementors, such as whether they will reduce or increase their investments in the platform due to the perception of unfair treatment or in the hope of preferential treatment. Future research can be guided by the question, "How do outcomes affect and reshape platform owners' governance decisions?"

Finally, researchers may investigate the relationship between value co-creation and value capture because value capture may undermine a platform ecosystem's stability. Researchers have pointed out that if platform owners capture a large value or distribute it unfairly, complementors' recruitment can be hindered (Jacobides et al., 2018; Sterk et al., 2022). In the context of mobile platform-mediated networks, Oh et al. (2015) proposed a bargaining model to investigate how value is likely to be captured between platform providers and app developers. However, we have scarce knowledge regarding the optimal allocation of co-created value among actors, and this limitation can be explored in future research. Moreover, it is not always possible to understand the potential outcomes because complementors adjust their value co-creation and capture strategies to respond to governance choices (Wen & Zhu, 2019). Thus, future research could focus on the dynamics of value co-creation and capture among actors during the evolution of a platform ecosystem.

Overall, platform ecosystem governance is not a one-shot decision; rather, it is dynamic and must be revised over time (Gawer, 2014; Wareham et al., 2014). Therefore, researchers may rely on longitudinal studies to investigate the topic in the suggested directions.

5.2. Practical implications

Despite the fact that this review was conducted with researchers as the main audience, the findings can be translated into managerial insights for practitioners who are willing to govern their digital platform ecosystems from the building phase to a mature one. The conceptual model lays out which components practitioners must consider to govern digital platform ecosystems in a more structured manner. Moreover, the proposed model suggests how these components are interrelated and affect one another. The three main components are strictly related and are narrowed down in a funnel, from decisions of structure, down to the mechanisms and further on to scope. When establishing a digital platform ecosystem, practitioners may initially approach these three components in a sequential way. These decisions are influenced by and must align with the context (e.g., the characteristics of the platform ecosystem, its actors, and the market) to be able to realize the intended outcome(s). Over time, given the inherently fragile nature of governance choices, managers should continuously adjust those choices while dealing with new and emergent issues (Foss et al., 2023). Subsequent choices may not follow a sequential approach; for instance, practitioners may broaden the governance scope without changing the governance mechanisms. However, practitioners should also be aware that their previous choices, capabilities, and bargaining power may affect the possibility of changing those previous governance choices, as well as their value co-creation and capture opportunities (Helfat & Raubitschek, 2018; Huber et al., 2017; Uzunca et al., 2022).

Overall, the conceptual model can be used by practitioners as a template for building and sustaining ecosystems by considering the different components, analyzing the current situation, monitoring changes, taking more informed governance choices, and adjusting them, if needed.

6. Conclusion and limitations

This multidisciplinary, systematic literature review synthesizes the extant knowledge on platform ecosystem governance. This review was guided by two research questions: "What are the building blocks of platform ecosystem governance?" and "What do we need more research on?" The answer to the first question is provided through a conceptual model of platform governance consisting of five building blocks: governance structure, governance mechanisms, governance scope, contingency factors, and outcomes. The answer to the second question is provided through the research agenda, which brings up several questions within each of the building blocks as promising avenues for future research.

This review is mainly helpful for academics. Specifically, by grouping concepts into blocks and suggesting how these blocks are related—which has not been addressed by extant research—the developed model presents a robust foundation for researchers investigating the topic of digital platform ecosystem governance for the first time. In addition, the conceptual model provides a basis for future research. Researchers may investigate the relationships among the different building blocks (or also within) of digital platform ecosystem governance through qualitative or quantitative studies.

The answer to the second question is a research agenda that derives from existing research, identified research gaps, and the evolving platform ecosystem scenario. Based on the proposed model, the research agenda suggests future research avenues and questions in each of the building blocks of digital platform ecosystem governance to enrich our knowledge of this vital topic.

Apart from motivating researchers to enhance our understanding of platform ecosystem governance, this review can also assist practitioners. In fact, the conceptual model draws attention to the building blocks and their interactions, which must be considered in building and/or sustaining a platform ecosystem in a structured manner.

Finally, this literature review follows the recommendations prescribed by Fink (2013), Okoli and Schabram (2010), and Webster and Watson (2002). Compliance with these guidelines and a focus on high-quality research give confidence in the thoroughness of the method and the identification of relevant contributions. However, some limitations still exist regarding subjective choices related to the selected keywords, the inclusion and exclusion criteria, and the time frame.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

*included in the literature analysis

Adam, M., Croitor, E., Werner, D., Benlian, A., & Wiener, M. (2023). Input control and its signalling effects for complementors' intention to join digital platforms. *Information Systems Journal, 33*(3), 437–466. https://doi.org/10.1111/isj.12408

Adner, R., & Kapoor, R. (2010). Value creation in innovation ecosystems: How the structure of technological interdependence affects firm performance in new technology generations. *Strategic Management Journal, 31*(3), 306–333. https://www.jstor.org/stable/40587479.

* Askay, D. (2017). A conceptual framework for investigating organizational control and resistance in crowd-based platforms. In T. Bui (Ed.), *Proceedings of the 50th Hawaii international conference on system sciences (HICSS)* (pp. 1727–1736). AIS Electronic Library (AISeL). https://doi.org/10.24251/HICSS.2017.210.

* Benlian, A., Hilkert, D., & Hess, T. (2015). How open is this platform? The meaning and measurement of platform openness from the complementers' perspective. *Journal of Information Technology, 30*(3), 209–228. https://doi.org/10.1057/jit.2015.6.

* Brandwijk, N., & de Reuver, M. (2023). Do negative externalities affect platform membership conditions? The impact of moral and regulatory legitimacy on platform governance. In M. Aanestad, S. Klein, & M. Tarafdar (Eds.), *Thirty-first European conference on information systems (ECIS 2023)* (pp. 1–17). Association for Information Systems (AIS) https://aisel.aisnet.org/ecis2023_rp/313.

* Burda, M., Locca, M., & Staykova, K. (2022). Decision rights decentralization in DeFi platforms. In R. Beck, D. Petcu, & M. Fotache (Eds.), *Thirtieth European conference on information systems (ECIS 2022)* (pp. 1–15). Association for Information Systems (AIS) https://aisel.aisnet.org/ecis2022_rp/145.

* Burtch, G., He, Q., Hong, Y., & Lee, D. D. (2019). Peer recognition increases user content generation but reduces content novelty. In H. Krcmar, J. Fedorowicz, W. Fong Boh, J. M. Leimeister, & S. Wattal (Eds.), *Fortieth international conference on information systems (ICIS)* (pp. 1–17). Association for Information Systems (AIS). https://doi.org/10.2139/ssrn.3465879.

* Cenamor, J., & Frishammar, J. (2021). Openness in platform ecosystems: Innovation strategies for complementary products. *Research Policy, 50*(1). https://doi.org/10.1016/j.respol.2020.104148.

* Cennamo, C., & Santaló, J. (2019). Generativity tension and value creation in platform ecosystems. *Organization Science, 30*(3), 617–641. https://doi.org/10.1287/orsc.2018.1270.

* Chen, J., Luo, J., Hu, W., & Ma, J. (2023-a). Fit into work! From formalizing governance of gig platform ecosystems to helping gig workers craft their platform work. *Decision Support Systems*. https://doi.org/10.1016/j.dss.2023.114016 (article (in press).

* Claussen, J., Kretschmer, T., & Mayrhofer, P. (2013). The effects of rewarding user engagement: The case of Facebook apps. *Information Systems Research, 24*(1), 186–200. https://doi.org/10.1287/isre.1120.0467.

* Constantinides, P., Parker, G., & Henfridsson, O. (2018). Platforms and infrastructures in the digital age. *Information Systems Research, 29*(2), 1–20. https://doi.org/10.1287/isre.2018.0794.

* Croitor, E., & Adam, M. (2020). Perceived input control on digital platforms: An empirical investigation. In F. Rowe, R. El Amrani, M. Limayem, S. Newell, N. Pouloudi, E. van Heck, & A. El Quammah (Eds.), *Proceedings of twenty-eighth European conference on information systems (ECIS)*. Association for Information Systems (AIS) https://aisel.aisnet.org/ecis2020_rp/2.

* Croitor, E., & Werner, D. (2021). Exploring the relationship between perceived input control and complementors' perceived performance: An empirical study on Amazon. *ECIS 2021 Research Papers, 13* https://aisel.aisnet.org/ecis2021_rp/13.

* Croitor, E., Werner, D., & Benlian, A. (2021). The effects of control mechanisms on complementors' behavioral intentions: An empirical study of reward-based crowdfunding platforms. In T. Bui (Ed.), *Proceedings of the 54th Hawaii international conference on system sciences (HICSS)* (pp. 6173–6182). University of Hawaii http://hdl.handle.net/10125/71365.

* Dellermann, D., & Reck, F. (2017). The antecedents of transaction costs in digital ecosystems: A configurational view on the interplay of app architecture and platform governance. In T. Bui (Ed.), *Proceedings of the 50th Hawaii international conference on system sciences (HICSS)* (pp. 5265–5274). AIS Electronic Library (AISeL). https://doi.org/10.24251/HICSS.2017.638.

* Eaton, B., Elaluf-Calderwood, S., Sørensen, C., & Yoo, Y. (2015). Distributed tuning of boundary resources. *MIS Quarterly, 39*(1), 217–244 https://www.jstor.org/stable/26628348.

* Engert, M., Evers, J., Hein, A., & Krcmar, H. (2022). The engagement of complementors and the role of platform boundary resources in e-commerce platform ecosystems. *Information Systems Frontiers, 24*(6), 2007–2025. https://doi.org/10.1007/s10796-021-10236-3.

* Engert, M., Evers, J., Hein, A., & Krcmar, H. (2023). Sustaining complementor engagement in digital platform ecosystems: Antecedents, behaviours and engagement trajectories. *Information Systems Journal, 33*, 1151–1185. https://doi.org/10.1111/isj.12438.

* Ens, N., Hukal, P., & Blegind Jensen, T. (2023). Dynamics of control on digital platforms. *Information Systems Journal, 33*, 890–911. https://doi.org/10.1111/isj.12429.

Fink, A. (2013). *Conducting research literature reviews: From the internet to paper* (4th ed.). Sage Publications.

Floetgen, R. J., Novotny, M., Hein, A., Weking, J., Urmetzer, F., Böhm, M., & Krcmar, H. (2022). Digital platform ecosystem performance: Antecedents and interrelations. In *Proceedings of the Australasian conference on information systems (ACIS)* (pp. 1–11). Association for Information Systems (AIS). https://acis.aaisnet.org/wp-content/uploads/2022/11/ACIS_2022_paper_4.pdf.

* Floetgen, R. J., Winder, P., Fielt, E., Weking, J., Hein, A., Schreieck, M., & Krcmar, H. (2023). Ownership structures and governance strategies for digital platform ecosystems: An empirical study. In M. Aanestad, S. Klein, & M. Tarafdar (Eds.), *Thirty-first European conference on information systems (ECIS 2023)* (pp. 1–18). Association for Information Systems (AIS) https://aisel.aisnet.org/ecis2023_rp/246.

* Foerderer, J., Kude, T., Mithas, S., & Heinzl, A. (2018). Does platform owner's entry crowd out innovation? Evidence from Google photos. *Information Systems Research, 29*(2), 444–460. https://doi.org/10.1287/isre.2018.0787.

* Foerderer, J., Kude, T., Schuetz, S. W., & Heinzl, A. (2019). Knowledge boundaries in enterprise software platform development: Antecedents and consequences for platform governance. *Information Systems Journal, 29*(1), 119–144. https://doi.org/10.1111/isj.12186.

* Foerderer, J., Lueker, N., & Heinzl, A. (2021). And the winner is…? The desirable and undesirable effects of platform awards. *Information Systems Research, 32*(4), 1155–1172. https://doi.org/10.1287/isre.2021.1019.

* Foss, N. J., Schmidt, J., & Teece, D. J. (2023). Ecosystem leadership as a dynamic capability. *Long Range Planning, 56*(1), 1–16. https://doi.org/10.1016/j.lrp.2022.102270, 102270.

* Gawer, A. (2014). Bridging differing perspectives on technological platforms: Toward an integrative framework. *Research Policy, 43*(7), 1239–1249. https://doi.org/10.1016/j.respol.2014.03.006.

* Gawer, A., & Cusumano, M. A. (2014). Industry platforms and ecosystem innovation. *Journal of Product Innovation Management, 31*(3), 417–433. https://doi.org/10.1111/jpim.12105.

* Ghazawneh, A., & Henfridsson, O. (2010). Governing third-party development through platform boundary resources. In R. Sabherwal, & M. Sumner (Eds.), *Thirty-first international conference on information systems (ICIS)* (pp. 1–17). Association for Information Systems (AIS) https://aisel.aisnet.org/icis2010_submissions/48.

* Ghazawneh, A., & Henfridsson, O. (2011). Micro-strategizing in platform ecosystems: A multiple case study. In D. F. Galletta, & T. P. Liang (Eds.), *Thirty-second international conference on information systems (ICIS)* (pp. 1–19). Association for Information Systems (AIS) https://aisel.aisnet.org/icis2011/proceedings/generaltopics/3.

* Ghazawneh, A., & Henfridsson, O. (2013). Balancing platform control and external contribution in third-party development: The boundary resources model. *Information Systems Journal, 23*(2), 173–192. https://doi.org/10.1111/j.1365-2575.2012.00406.x.

* Glaser, K. (2020). Governance of internal complementors in platform ecosystems. In J. F. George, S. Paul, R. De, E. Karahanna, S. Sarker, & G. Oestreicher-Singer (Eds.), *Forty-first international conference on information systems (ICIS)* (pp. 1–19). Association for Information Systems (AIS) https://aisel.aisnet.org/icis2020/is_development/is_development/10.

* Gol, E. S., Avital, M., & Stein, M. K. (2019a). Crowdwork platforms: Juxtaposing centralized and decentralized governance. In P. Johannesson, P. Ågerfalk, & R. Helms (Eds.), *Proceedings of the 27th European conference on information systems (ECIS)* (pp. 1–16). Association for Information Systems (AIS) https://aisel.aisnet.org/ecis2019_rp/110.

* Gol, E. S., Stein, M. K., & Avital, M. (2019b). Crowdwork platform governance toward organizational value creation. *The Journal of Strategic Information Systems, 28*(2), 175–195. https://doi.org/10.1016/j.jsis.2019.01.001.

* Goldbach, T., & Benlian, A. (2015a). How social capital facilitates clan control on software platforms to enhance App-developers' performance and success. In T. A. Carte, A. Heinzl, & C. Urquhart (Eds.), *Thirty-sixth international conference on information systems (ICIS)* (pp. 1–18). Association for Information Systems (AIS) https://aisel.aisnet.org/icis2015/proceedings/ISgov/2.

* Goldbach, T., & Benlian, A. (2015b). Understanding informal control modes on software platforms-The mediating role of third-party developers' intrinsic motivation. In T. A. Carte, A. Heinzl, & C. Urquhart (Eds.), *Thirty-Sixth international conference on information systems (ICIS* (pp. 1–19). Association for Information Systems (AIS) https://aisel.aisnet.org/icis2015/proceedings/ManagingIS/2.

* Goldbach, T., Benlian, A., & Buxmann, P. (2018). Differential effects of formal and self-control in mobile platform ecosystems: Multi-method findings on third-party

developers' continuance intentions and application quality. *Information & Management, 55*(3), 271–284. https://doi.org/10.1016/j.im.2017.07.003.
* Goldbach, T., & Kemper, V. (2014). Should I stay or should I go? The effects of control mechanisms on app developers' intention to stick with a platform. In M. Avital, J. M. Leimeister, & U. Schultze (Eds.), *Proceedings of the European conference on information systems (ECIS)* (pp. 1–15). Association for Information Systems (AIS) http://aisel.aisnet.org/ecis2014/proceedings/track15/15.
* Goldbach, T., Kemper, V., & Benlian, A. (2014). Mobile application quality and platform stickiness under formal vs. self-control—Evidence from an experimental study. In M. D. Myers, & D. W. Straub (Eds.), *Thirty-fifth international conference on information systems (ICIS)* (pp. 1–18). Association for Information Systems (AIS) https://aisel.aisnet.org/icis2014/proceedings/ProjectManagement/7.
* Göttel, V. (2021). Sharing algorithmic management information in the sharing economy–effects on workers' intention to stay with the organization. In J. S. Valacich, A. Barua, R. Wright, A. Kankanhalli, X. Li, & S. Miranda (Eds.), *Forty-second international conference on information systems (ICIS)* (pp. 1–17). Association for Information Systems (AIS) https://aisel.aisnet.org/icis2021/sharing_econ/sharing_econ/3.
Gulati, R., Puranam, P., & Tushman, M. (2012). Meta-organization design: Rethinking design in interorganizational and community contexts. *Strategic Management Journal, 33*(6), 571–586. https://doi.org/10.1002/smj.1975
* Guo, W., Straub, D., Zhang, P., & Cai, Z. (2021). How trust lead to commitment on microsourcing platforms: Unraveling the effects of governance and third-party mechanisms on triadic microsourcing relationships. *MIS Quarterly, 45*(3), 1309–1348. https://doi.org/10.25300/MISQ/2021/14349.
* Gutt, D., Neumann, J., Jabr, W., & Kundisch, D. (2019). The app updating conundrum: Implications of platform's rating resetting on developers' behavior. In H. Krcmar, J. Fedorowicz, W. Fong Boh, J. M. Leimeister, & S. Wattal (Eds.), *Fortieth international conference on information systems* (pp. 1–17). Association for Information Systems (AIS) https://aisel.aisnet.org/icis2019/mobile_iot/mobile_iot/5.
* Halckenhaeusser, A., Foerderer, J., & Heinzl, A. (2020). Platform governance mechanisms: An integrated literature review and research directions. In F. Rowe, R. El Amrani, M. Limayem, S. Newell, N. Pouloudi, E. van Heck, & A. El Quammah (Eds.), *Proceedings of the 28th European conference on information systems (ECIS)* (pp. 15–17). Association for Information Systems (AIS) https://aisel.aisnet.org/ecis2020_rp/30.
* He, S., Peng, J., Li, J., & Xu, L. (2020). Impact of platform owner's entry on third-party stores. *Information Systems Research, 31*(4), 1467–1484. https://doi.org/10.1287/isre.2020.0957.
* Heimburg, V., & Wiesche, M. (2022). Relations between actors in digital platform ecosystems: A literature review. In R. Beck, D. Petcu, & M. Fotache (Eds.), *Thirtieth European conference on information systems (ECIS 2022)* (pp. 1–21). Association for Information Systems (AIS) https://aisel.aisnet.org/ecis2022_rp/93.
Hein, A., Schreieck, M., Riasanow, T., Setzke, D. S., Wiesche, M., Böhm, M., & Krcmar, H. (2020). Digital platform ecosystems. *Electronic Markets, 30*(1), 87–98. https://doi.org/10.1007/s12525-019-00377-4
Hein, A., Schreieck, M., Wiesche, M., & Krcmar, H. (2016). Multiple-case analysis on governance mechanisms of multi-sided platforms. In *Multikonferenz wirtschaftsinformatik [Multi-conference on business informatics* (pp. 9–11). Universitätsverlag Ilmenau [Ilmenau University Press.
* Helfat, C. E., & Raubitschek, R. S. (2018). Dynamic and integrative capabilities for profiting from innovation in digital platform-based ecosystems. *Research Policy, 47*(8), 1391–1399. https://doi.org/10.1016/j.respol.2018.01.019.
* Hilkert, D., Benlian, A., Sarstedt, M., & Hess, T. (2011). Perceived software platform openness: The scale and its impact on developer satisfaction. In D. F. Galletta, & T. P. Liang (Eds.), *Thirty-second international conference on information systems (ICIS)* (pp. 1–20). Association for Information Systems (AIS) https://aisel.aisnet.org/icis2011/proceedings/onlinecommunity/13.
* Hirsch, F., Alizadeh, A., Wiener, M., & Cram, A. W. (2023). Algorithmic control in platform and traditional work settings: An updated conceptual framework. In M. Aanestad, S. Klein, & M. Tarafdar (Eds.), *Thirty-first European conference on information systems (ECIS 2023)* (pp. 1–16). Association for Information Systems (AIS) https://aisel.aisnet.org/ecis2023_rp/247.
* Huber, T. L., Kude, T., & Dibbern, J. (2017). Governance practices in platform ecosystems: Navigating tensions between cocreated value and governance costs. *Information Systems Research, 28*(3), 563–584. https://doi.org/10.1287/isre.2017.0701.
* Hurni, T., & Huber, T. (2014). The interplay of power and trust in platform ecosystems of the enterprise application software industry. In M. Avital, J. M. Leimeister, & U. Schultze (Eds.), *Proceedings of the Twenty-second European conference on information systems (ECIS)* (pp. 1–15). Association for Information Systems (AIS) https://aisel.aisnet.org/ecis2014/proceedings/track15/19/.
* Hurni, T., Huber, T. L., & Dibbern, J. (2022). Power dynamics in software platform ecosystems. *Information Systems Journal, 32*(2), 310–343. https://doi.org/10.1111/isj.12356.
* Hurni, T., Huber, T. L., Dibbern, J., & Krancher, O. (2021). Complementor dedication in platform ecosystems: Rule adequacy and the moderating role of flexible and benevolent practices. *European Journal of Information Systems, 30*(3), 237–260. https://doi.org/10.1080/0960085X.2020.1779621.
Iansiti, M., & Levien, R. (2004). Strategy as ecology. *Harvard Business Review, 82*, 68–81.
* Ingram Bogusz, C., & Kashyap, S. (2022). The carrot and the stick: Open banking platform governance in the shadow of PSD2. In R. Beck, D. Petcu, & M. Fotache (Eds.), *Thirtieth European conference on information systems (ECIS 2022)* (pp. 1–14). Association for Information Systems (AIS) https://aisel.aisnet.org/ecis2022_rp/113.
* Inoue, Y. (2021). Indirect innovation management by platform ecosystem governance and positioning: Toward collective ambidexterity in the ecosystems. *Technological Forecasting and Social Change, 166*(120652), 1–15. https://doi.org/10.1016/j.techfore.2021.120652.
* Islam, H. A., Farrell, M., Nair, A., & Zhang, J. (2023). Understanding transaction platform governance and conflicts: A configuration approach. *Technological Forecasting and Social Change, 189*(122382), 1–12. https://doi.org/10.1016/j.techfore.2023.122382.
* Jacobides, M. G., Cennamo, C., & Gawer, A. (2018). Towards a theory of ecosystems. *Strategic Management Journal, 39*(8), 2255–2276. https://doi.org/10.1002/smj.2904.
* Jiang, J. (2023). Covert resistance against algorithmic control of online labor platforms – a systematic literature review. In M. Aanestad, S. Klein, & M. Tarafdar (Eds.), *Thirty-first European conference on information systems (ECIS 2023)* (pp. 1–17). Association for Information Systems (AIS) https://aisel.aisnet.org/ecis2023_rp/272.
* Jovanovic, M., Kostić, N., Sebastian, I. M., & Sedej, T. (2022a). Managing a blockchain-based platform ecosystem for industry-wide adoption: The case of TradeLens. *Technological Forecasting and Social Change, 184*(121981), 1–14. https://doi.org/10.1016/j.technovation.2020.102218.
* Jovanovic, M., Sjödin, D., & Parida, V. (2022b). Co-evolution of platform architecture, platform services, and platform governance: Expanding the platform value of industrial digital platforms. *Technovation, 118*(102218), 1–14. https://doi.org/10.1016/j.technovation.2020.102218.
* Kindermann, B., Salge, T. O., Wentzel, D., Flatten, T. C., & Antons, D. (2022). Dynamic capabilities for orchestrating digital innovation ecosystems: Conceptual integration and research opportunities. *Information and Organization, 32*(3), 1–16. https://doi.org/10.1016/j.infoandorg.2022.100422, 100422.
* Koo, W. W., & Eesley, C. E. (2021). Platform governance and the rural–urban divide: Sellers' responses to design change. *Strategic Management Journal, 42*(5), 941–967. https://doi.org/10.1002/smj.3259.
Kretschmer, T., & Claussen, J. (2016). Generational transitions in platform markets – the role of backward compatibility. *Strategy Science, 1*(2), 90–104. https://doi.org/10.1287/stsc.2015.0009
* Kretschmer, T., Leiponen, A., Schilling, M., & Vasudeva, G. (2022). Platform ecosystems as meta-organizations: Implications for platform strategies. *Strategic Management Journal, 43*(3), 405–424. https://doi.org/10.1002/smj.3250.
* de Lima Fontão, A., dos Santos, R. P., & Dias-Neto, A. C. (2019). Exploiting repositories in mobile software ecosystems from a governance perspective. *Information Systems Frontiers, 21*(1), 143–161. https://doi.org/10.1007/s10796-018-9861-8.
* Lin, F., Tian, H., Zhao, J., & Chi, M. (2023). Reward or punish: Investigating output controls and content generation in the multi-sided platform context. *Internet Research, 33*(2), 578–605. https://doi.org/10.1108/INTR-05-2021-0292.
* Luo, D., Fang, Y., Song, P., & Wang, C. (2019). The impact of digital platform rapid release strategy on app update behavior: An empirical study of Firefox. In H. Krcmar, J. Fedorowicz, W. Fong Boh, J. M. Leimeister, & S. Wattal (Eds.), *40th International conference on information systems (ICIS 2019)* (p. 1807). Association for Information Systems (AIS) https://aisel.aisnet.org/icis2019/is_heart_of_innovation_ecosystems/innovation_ecosystems/7.
* Mei, J., Zheng, G., & Zhu, L. (2022). Governance mechanisms implementation in the evolution of digital platforms: A case study of the Internet of Things platform. *R&D Management, 52*(3), 498–516. https://doi.org/10.1111/radm.12494.
Miles, M. B., & Huberman, A. M. (1994). *Qualitative data analysis: An expanded sourcebook* (2nd ed.). SAGE Publications.
* Miric, M., Ozalp, H., & Yilmaz, E. D. (2023). Trade-offs to using standardized tools: Innovation enablers or creativity constraints?. *Strategic Management Journal, 44*(4), 909–942. https://doi.org/10.1002/smj.3457.
* Mohagheghzadeh, A., & Svahn, F. (2016a). Shifting design capability to third-party developers: An affordance perspective on platform boundary resources. In *Twenty-second Americas conference on information systems (AMCIS)* (pp. 1–10). Association for Information Systems (AIS) https://aisel.aisnet.org/amcis2016/Open/Presentations/8.
* Mohagheghzadeh, A., & Svahn, F. (2016b). Transforming organizational resource into platform boundary resources. In *ECIS proceedings* (Vol. 9)Research Papers http://aisel.aisnet.org/ecis2016_rp/9.
* Möhlmann, M., Zalmanson, L., Henfridsson, O., & Gregory, R. W. (2021). Algorithmic management of work on online labor platforms: When matching meets control. *MIS Quarterly, 45*(4), 1999–2022. https://doi.org/10.25300/MISQ/2021/15333.
Mukhopadhyay, S., & Bouwman, H. (2019). Orchestration and governance in digital platform ecosystems: A literature review and trends. *Digital Policy, Regulation and Governance, 21*(4), 329–351. https://doi.org/10.1108/DPRG-11-2018-0067
* Niedermayer, A. (2013). On platforms, incomplete contracts, and open source software. *International Journal of Industrial Organization, 31*(6), 714–722. https://doi.org/10.1016/j.ijindorg.2013.07.003.
Oh, J., Koh, B., & Raghunathan, S. (2015). Value appropriation between the platform provider and app developers in mobile platform mediated networks. *Journal of Information technology, 30*, 245–259. https://doi.org/10.1057/jit.2015.21
Okoli, C., & Schabram, K. (2010). A guide to conducting a systematic literature review of information systems research. *Social Science Research Network, 10*(26), 1–49. https://doi.org/10.2139/ssrn.1954824
* O'Mahony, S., & Karp, R. (2022). From proprietary to collective governance: How do platform participation strategies evolve?. *Strategic Management Journal, 43*(3), 530–562. https://doi.org/10.1002/smj.3150.
Pawson, R., Tilley, N., & Tilley, N. (1997). *Realistic evaluation*. Sage Publishing.
* Pereira, J., Tavalaei, M. M., & Ozalp, H. (2019). Blockchain-based platforms: Decentralized infrastructures and its boundary conditions. *Technological Forecasting and Social Change, 146*, 94–102. https://doi.org/10.1016/j.techfore.2019.04.030.
* Perscheid, G., Ostern, N. K., & Moormann, J. (2020). Determining platform governance: Framework for classifying governance types. In J. F. George, S. Paul, R. De, E. Karahanna, S. Sarker, & G. Oestreicher-Singer (Eds.), *Forty-First international*

conference on information systems (ICIS) (pp. 1–16). Association for Information Systems (AIS) https://aisel.aisnet.org/icis2020/governance_is/governance_is/8.

Pidun, U., Reeves, M., & Schüssler, M. (2020). *Why do most business ecosystems fail?* Boston Consulting Group. https://www.bcg.com/publications/2020/why-do-most-business-ecosystems-fail. (Accessed 20 September 2023).

Rainey, H. G., Backoff, R. W., & Levine, C. H. (1976). Comparing public and private organizations. *Public Administration Review, 36*(2), 233–244. https://doi.org/10.2307/975145

Reeves, M., Lotan, H., Legrand, J., & Jacobides, M. G. (2019). How business ecosystems rise (and often fall). *MIT Sloan Management Review, 60*(4), 1–6. https://sloanreview.mit.edu/article/how-business-ecosystems-rise-and-often-fall/.

de Reuver, M., Sørensen, C., & Basole, R. C. (2018). The digital platform: A research agenda. *Journal of Information Technology, 33*(2), 124–135. https://doi.org/10.1057/s41265-016-0033-3

* Riasanow, T., Burckhardt, F., Soto Setzke, D., Böhm, M., & Krcmar, H. (2018). The generic blockchain ecosystem and its strategic implications. In *Twenty-fourth Americas conference on information systems (AMCIS)* (pp. 1–10). Association for Information Systems (AIS) https://aisel.aisnet.org/amcis2018/StrategicIT/Presentations/13.

* Rickmann, T., Wenzel, S., & Fischbach, K. (2014). Software ecosystem orchestration: The perspective of complementors. In *Twentieth Americas conference on information systems* (pp. 1–14). Association for Information Systems (AIS) https://aisel.aisnet.org/amcis2014/e-Business/GeneralPresentations/7.

Rietveld, J., & Schilling, M. A. (2021). Platform competition: A systematic and interdisciplinary review of the literature. *Journal of Management, 47*(6), 1528–1563. https://doi.org/10.1177/0149206320969791

* Rietveld, J., Schilling, M. A., & Bellavitis, C. (2019). Platform strategy: Managing ecosystem value through selective promotion of complements. *Organization Science, 30*(6), 1232–1251. https://doi.org/10.1287/orsc.2019.1290.

* Saadatmand, F., Lindgren, R., & Schultze, U. (2019). Configurations of platform organizations: Implications for complementor engagement. *Research Policy, 48*(8). https://doi.org/10.1016/j.respol.2019.03.015.

* Schreieck, M., & Wiesche, M. (2017). How established companies leverage IT platforms for value co-creation–insights from banking. In I. Ramos, V. Tuunainen, & H. Krcmar (Eds.), *Proceedings of the 25th European conference on information systems (ECIS)* (pp. 1726–1741). Association for Information Systems (AIS) https://aisel.aisnet.org/ecis2017_rp/111.

* Schreieck, M., Wiesche, M., & Krcmar, H. (2016). Design and governance of platform ecosystems–key concepts and issues for future research. In *ECIS proceedings* (Vol. 76) Research Papers https://aisel.aisnet.org/ecis2016_rp/76.

* Schreieck, M., Wiesche, M., & Krcmar, H. (2021). Capabilities for value co-creation and value capture in emergent platform ecosystems: A longitudinal case study of SAP's cloud platform. *Journal of Information Technology, 36*(4), 365–390. https://doi.org/10.1177/02683962211023780.

* Schulze, L., Trenz, M., & Nickerson, R. C. (2021). Fingers in the pie: Characterizing decision rights partitioning on digital labor platforms. In J. S. Valacich, A. Barua, R. Wright, A. Kankanhalli, X. Li, & S. Miranda (Eds.), *Forty-second international conference on information systems (ICIS)* (pp. 1–17). Association for Information Systems (AIS) https://aisel.aisnet.org/icis2021/is_future_work/is_future_work/1.

* Selander, L., Henfridsson, O., & Svahn, F. (2013). Capability search and redeem across digital ecosystems. *Journal of Information Technology, 28*(3), 183–197. https://doi.org/10.1057/jit.2013.14.

* Shi, R., Aaltonen, A., Henfridsson, O., & Gopal, R. D. (2023). Comparing platform owners' early and late entry into complementary markets. *MIS Quarterly* (in press).

* Song, P., Xue, L., Rai, A., & Zhang, C. (2018). The ecosystem of software platform: A study of asymmetric cross-side network effects and platform governance. *MIS Quarterly, 42*(1), 121–142.

* Sterk, F., Heinz, D., Peukert, C., Fleuchaus, F., Kölbel, T., & Weinhardt, C. (2022). Fostering value co-creation in incumbent firms: The case of Bosch's IoT ecosystem landscape. In R. Beck, S. Petter, & T. Blegind Jensen (Eds.), *Forty-third international conference on information systems (ICIS)*. Association for Information Systems (AIS), 17 https://aisel.aisnet.org/icis2022/governance_is/governance_is/16.

* Suvivuo, S., Rinta-Kahila, T., & Tuunainen, V. K. (2023). Building and sustaining communities in online labour platforms. In M. Aanestad, S. Klein, & M. Tarafdar (Eds.), *Thirty-first European conference on information systems (ECIS 2023)* (pp. 1–15). Association for Information Systems (AIS) https://aisel.aisnet.org/ecis2023_rp/398.

* Tang, X., Wang, P., & Zhu, Z. (2021). Effects of centralized and distributed quality control on sales of experience goods in a transaction platform ecosystem. In *Twenty-seventh Americas conference on information systems (AMCIS)* (pp. 1–10). Association for Information Systems (AIS) https://aisel.aisnet.org/amcis2021/strategic_is/strategic_is/8.

* Thies, F., Wessel, M., & Benlian, A. (2018). Network effects on crowdfunding platforms: Exploring the implications of relaxing input control. *Information Systems Journal, 28*(6), 1239–1262. https://doi.org/10.1111/isj.12194.

Tiwana, A. (2013). *Platform ecosystems: Aligning architecture, governance, and strategy*. Elsevier Science.

* Tiwana, A. (2015). Evolutionary competition in platform ecosystems. *Information Systems Research, 26*(2), 266–281. https://doi.org/10.1287/isre.2015.0573.

* Tiwana, A., Konsynski, B., & Bush, A. A. (2010). Research commentary—platform evolution: Coevolution of platform architecture, governance, and environmental dynamics. *Information Systems Research, 21*(4), 675–687. https://doi.org/10.1287/isre.1100.0323.

Tiwana, A., Konsynski, B., & Venkatraman, N. (2013). Information technology and organizational governance: The IT governance cube. *Journal of Management Information Systems, 30*(3), 7–12. https://doi.org/10.2753/MIS0742-1222300301

* Uzunca, B., Sharapov, D., & Tee, R. (2022). Governance rigidity, industry evolution, and value capture in platform ecosystems. *Research Policy, 51*(7), 1–15. https://doi.org/10.1016/j.respol.2022.104560, 104560.

* Van Dyck, M., Lüttgens, D., Piller, F. T., Diener, K., & Pollok, P. (2021). Positioning strategies in emerging industrial ecosystems for industry 4.0: A longitudinal study of platform emergence in the agricultural industry. Universitätsbibliothek der RWTH Aachen. In T. Bui (Ed.), *Proceedings of the 54th Hawaii international conference on system sciences (HICSS)* (pp. 6153–6162). University of Hawaii http://hdl.handle.net/10125/71363.

* de Vasconcelos Gomes, L. A., Flechas, X. A., Facin, A. L. F., & Borini, F. M. (2021). Ecosystem management: Past achievements and future promises. *Technological Forecasting and Social Change, 171*, 1–15. https://doi.org/10.1016/j.techfore.2021.120950, 120950.

Vesselkov, A., Hämmäinen, H., & Töyli, J. (2019). Design and governance of mHealth data sharing. *Communications of the Association for Information Systems, 45*(1), 18. https://doi.org/10.17705/1CAIS.04518

* Wang, J. (2021). Do birds of a feather flock together? Platform's quality screening and end-users' choices theory and empirical study of online trading platforms. *International Journal of Industrial Organization, 75*(102698), 1–28. https://doi.org/10.1016/j.ijindorg.2020.102698.

* Wang, G., & Nandhakumar, J. (2017). Strategic swaying: How startups grow digital platforms. In Y. J. Kim, R. Agarwal, & J. K. Lee (Eds.), *Thirty Eighth international conference on information systems (ICIS)* (pp. 1–17). Association for Information Systems (AIS) https://aisel.aisnet.org/icis2017/Strategy/Presentations/17.

* Wareham, J., Fox, P. B., & Cano Giner, J. L. (2014). Technology ecosystem governance. *Organization Science, 25*(4), 1195–1215. https://doi.org/10.1287/orsc.2014.0895.

* Weber, M., Remus, U., & Pregenzer, M. (2022). A new era of control: Understanding algorithmic control in the gig economy. In R. Beck, S. Petter, & T. Blegind Jensen (Eds.), *Forty-third international conference on information systems (ICIS)*. Association for Information Systems (AIS), 17 https://aisel.aisnet.org/icis2022/is_futureofwork/is_futureofwork/1.

Webster, J., & Watson, R. T. (2002). Analyzing the past to prepare for the future: Writing a literature review. *MIS Quarterly, 26*. xiii–xxiii https://www.jstor.org/stable/4132319.

Weill, P., & Ross, J. W. (2004). *IT governance on one page*. SSRN Scholarly Paper No. ID 664612, Social Science Research Network https://papers.ssrn.com/sol3/papers.cfm?abstract_id=664612. (Accessed 20 July 2023).

* Weiss, N., Schreieck, M., Wiesche, M., & Krcmar, H. (2023). Lead complementor involvement in the design of platform boundary resources: A case study of BMW's onboard apps. *Information Systems Journal*, 1–33. https://doi.org/10.1111/isj.12449.

* Wen, W., & Zhu, F. (2019). Threat of platform-owner entry and complementor responses: Evidence from the mobile app market. *Strategic Management Journal, 40*(9), 1336–1367. https://doi.org/10.1002/smj.3031.

* Werner, J., Frost, S., & Zarnekow, R. (2020). Towards a taxonomy for governance mechanisms of blockchain-based platforms. In F. Rowe, R. El Amrani, M. Limayem, S. Newell, N. Pouloudi, E. van Heck, & A. El Quammah (Eds.). *ECIS proceedings* (Vol. 26). Research Papers https://aisel.aisnet.org/ecis2020_rp/26.

* Wessel, M., Thies, F., & Benlian, A. (2015). The effects of relinquishing control in platform ecosystems: Implications from a policy change on Kickstarter. In T. A. Carte, A. Heinzl, & C. Urquhart (Eds.), *Thirty-sixth international conference on information systems (ICIS)* (pp. 1–19). Association for Information Systems (AIS) https://aisel.aisnet.org/icis2015/proceedings/EconofIS/2.

* Wessel, M., Thies, F., & Benlian, A. (2017). Opening the floodgates: The implications of increasing platform openness in crowdfunding. *Journal of Information Technology, 32*(4), 344–360. https://doi.org/10.1057/s41265-017-0040-z.

* Wiener, M., Cram, W., & Benlian, A. (2021). Algorithmic control and gig workers: A legitimacy perspective of Uber drivers. *European Journal of Information Systems, 32*(3), 485–507. https://doi.org/10.1080/0960085X.2021.1977729.

* Zhang, Y., Li, J., & Tong, T. W. (2022). Platform governance matters: How platform gatekeeping affects knowledge sharing among complementors. *Strategic Management Journal, 43*(3), 599–626. https://doi.org/10.1002/smj.3191.

* Zhang, Y., Pinkse, J., & McMeekin, A. (2020). The governance practices of sharing platforms: Unpacking the interplay between social bonds and economic transactions. *Technological Forecasting and Social Change, 158*, 1–13. https://doi.org/10.1016/j.techfore.2020.120133, 120133.

* Zhu, F., & Liu, Q. (2018). Competing with complementors: An empirical look at Amazon.com. *Strategic Management Journal, 39*(10), 2618–2642. https://doi.org/10.1002/smj.2932.